

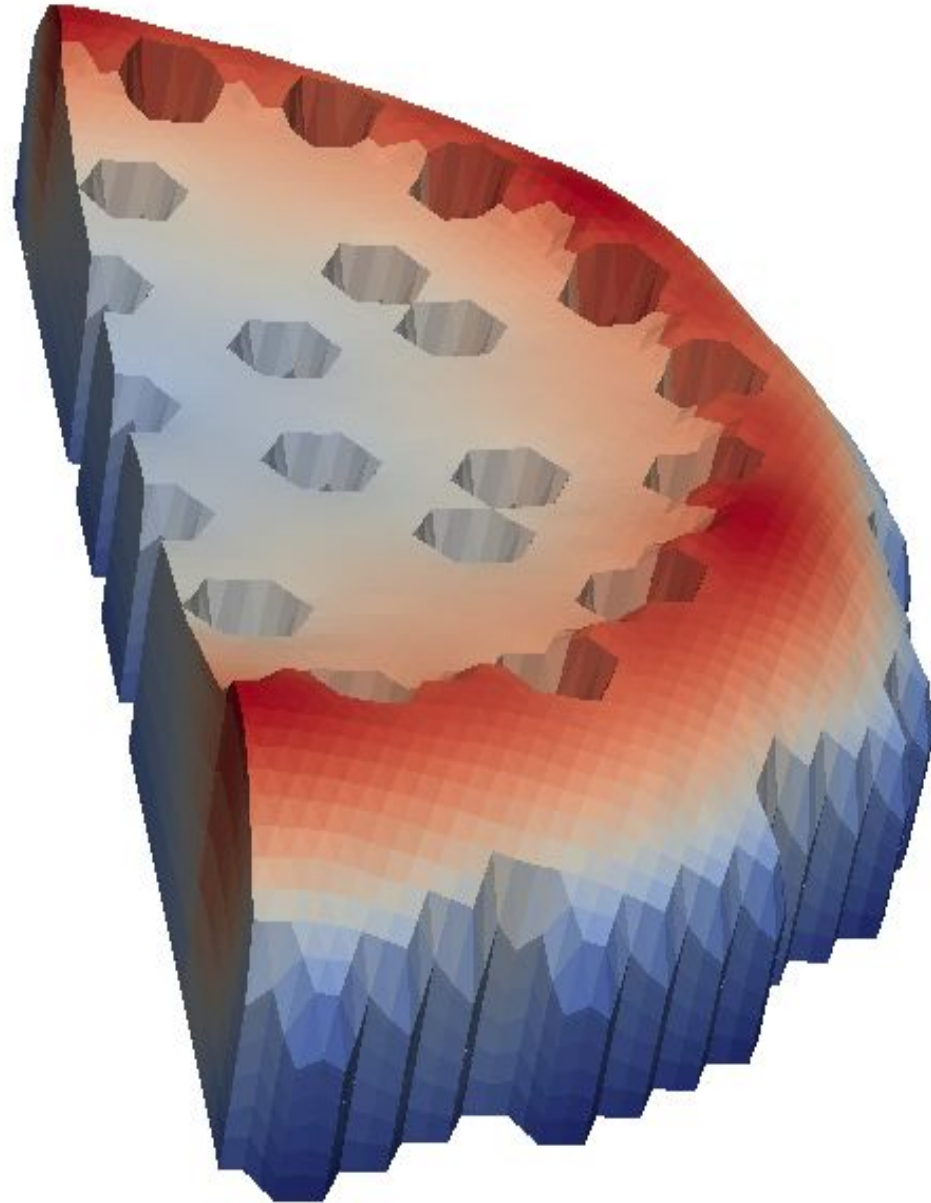


**Ecole Polytechnique
Fédérale de Lausanne**
EPFL

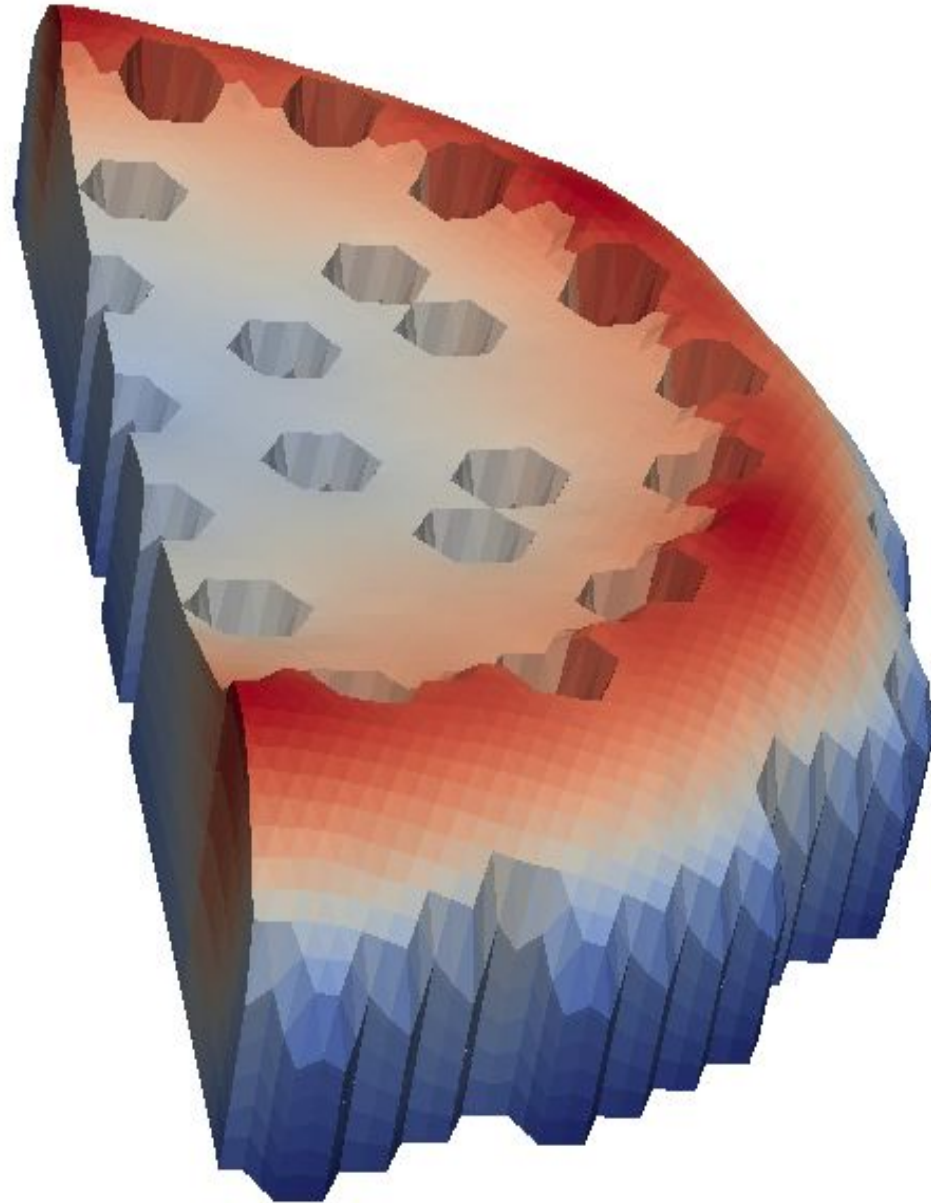
A short introduction to using GeN-Foam

Carlo Fiorina – carlo.fiorina@epfl.ch

- How to get it?
- How to install it?
- What's inside?
- Slack Forum
- An example
- The other tutorials



- How to get it?
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How to get it?

- **Free, online at**

- <https://gitlab.com/foam-for-nuclear/GeN-Foam/-/tree/develop>

- “Develop” branch or “Master” branch

- Either

- git clone

- <https://gitlab.com/foam-for-nuclear/GeN-Foam.git>

- or, simply download

- **Soon on the IAEA/ONCORE website:**

- <https://nucleus.iaea.org/sites/oncore/SitePages/Home.aspx>

How to get it?

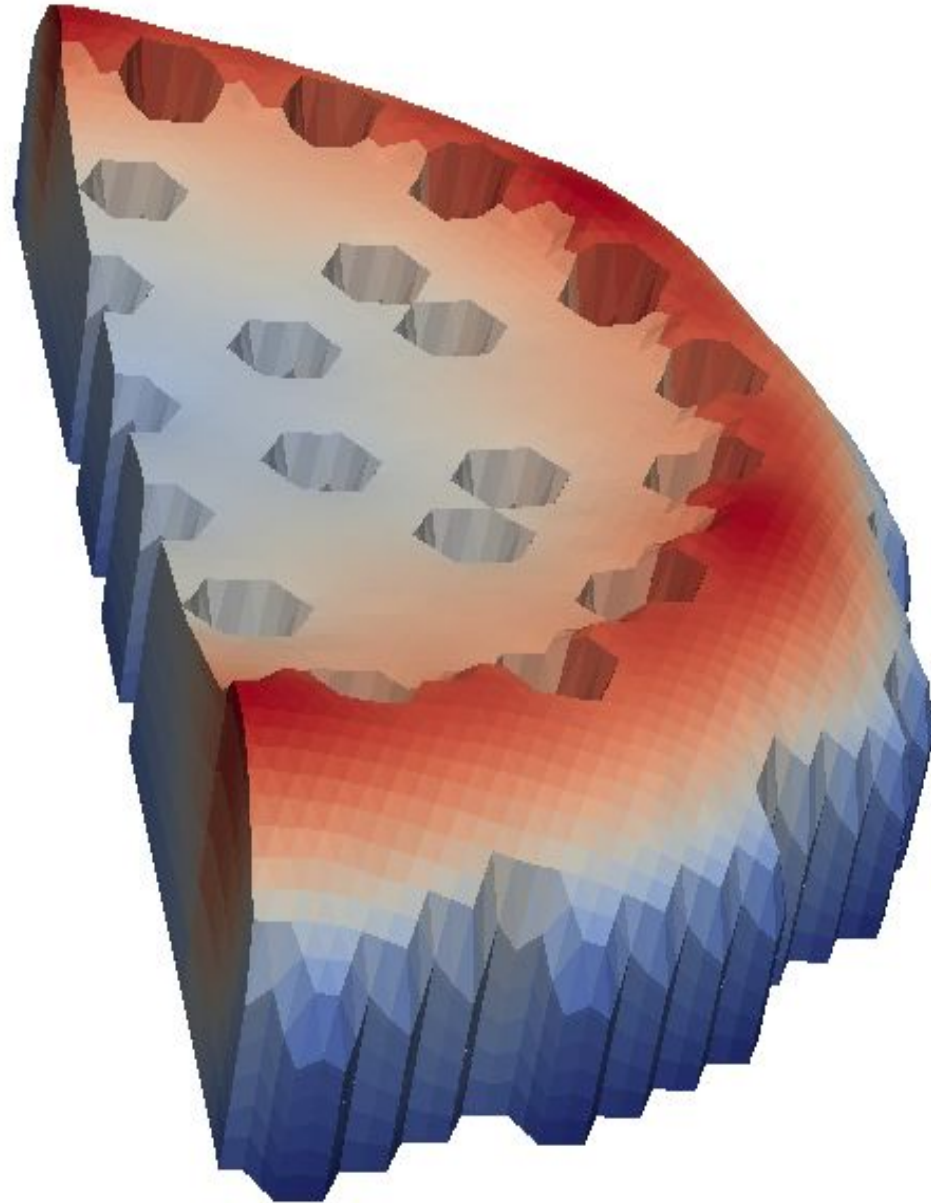
Download

Clone

The screenshot shows the GitLab interface for the 'GeN-Foam' repository. The left sidebar contains navigation links: Project overview, Repository, Files, Commits, Branches, Tags, Contributors, Graph, Compare, Locked Files, and Collapse sidebar. The main content area shows the repository path 'foam-for-nuclear project > GeN-Foam > Repository'. Below the path, there's a dropdown for 'develop' and a '+ v' button. To the right are buttons for 'History', 'Find file', 'Web IDE', and a download icon. A blue 'Clone' button is also present. Above the 'Clone' button, an arrow points from the word 'Clone' to the button, and another arrow points from the word 'Download' to the download icon. Below these buttons, a commit card shows the title 'Update solvePointKineticsLiquidFuel.H' by 'foam-for-nuclear project' from 22 hours ago, with a commit hash '0a05c5b4'. At the bottom, a table lists repository files and their last commit details.

Name	Last commit	Last update
Documentation	Deleted howTo file. Created README file in ...	9 months ago
GeN-Foam	Update solvePointKineticsLiquidFuel.H	22 hours ago
Tools	Resturetcured Tools folder	8 months ago
Tutorials	Corrected bug in the modifiedEngel fluid-str...	4 weeks ago
.gitignore	Added FFS library from my two-phase work t...	1 year ago

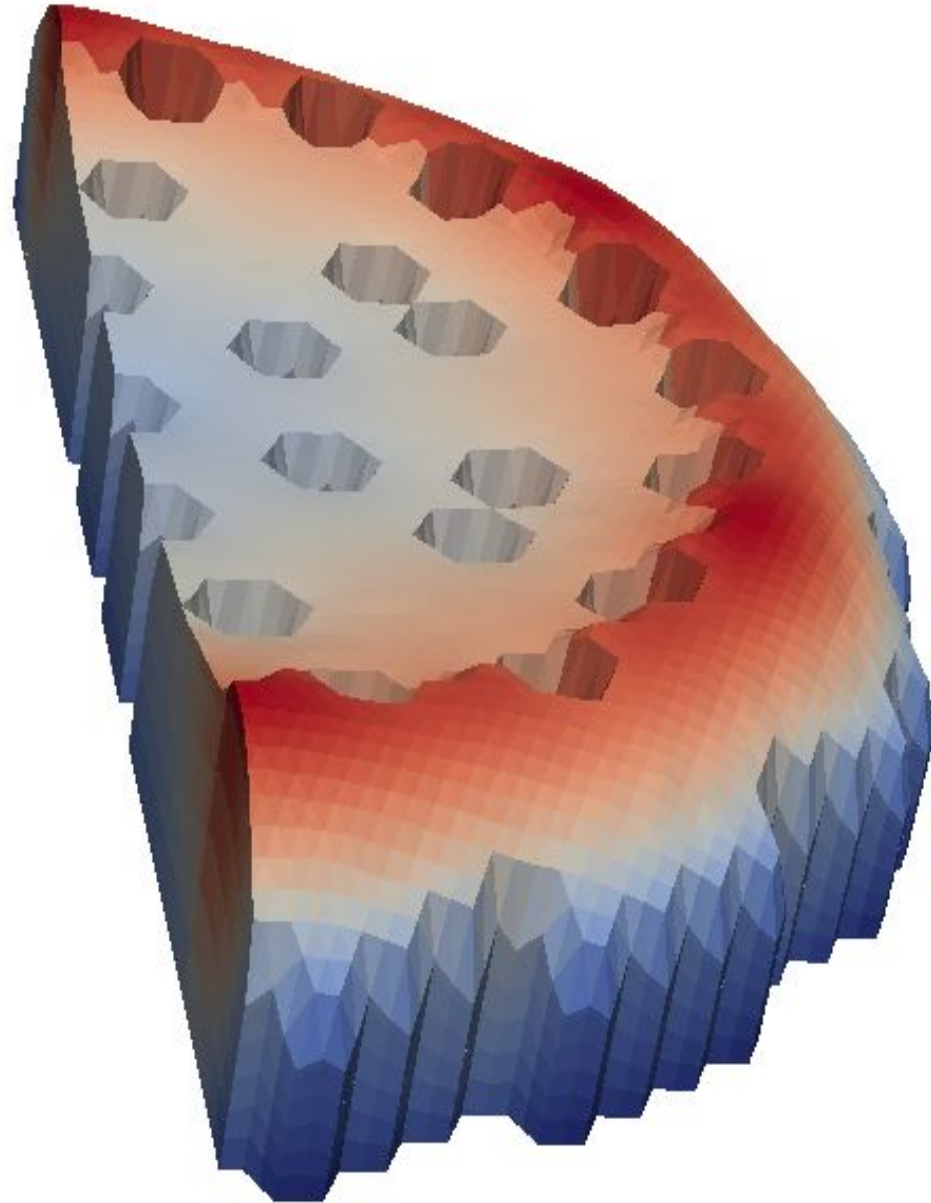
- How to get it?
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How to install it?

- Download OpenFOAM at
 - <https://www.openfoam.com/download/>
- Install OpenFOAM and prepare the environment
 - <https://www.openfoam.com/download/installation.php>
- Download GeN-Foam
- Enter the GeN-Foam /GeN-Foam folder and run:
 - *Allwclean*
 - *Allwmake* (or *Allwmake -j*, to compile in parallel)

- How to get it?
- How to install it?
- **What's inside?**
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What's inside?

develop

GeN-Foam / +

History

Find file

Web IDE



Clone

 Update solvePointKineticsLiquidFuel.H
foam-for-nuclear project authored 22 hours ago

0a05c5b4



Name	Last commit	Last update
 Documentation	Deleted howTo file. Created README file in ...	9 months ago
 GeN-Foam	Update solvePointKineticsLiquidFuel.H	22 hours ago
 Tools	Resturetcured Tools folder	8 months ago
 Tutorials	Corrected bug in the modifiedEngel fluid-str...	4 weeks ago
 .gitignore	Added FFS library from my two-phase work t...	1 year ago
 LICENSE	Add LICENSE file	3 months ago
 README	Update README	3 months ago

- README file often present to describe what's in a subfolder

What's inside? Source code

develop

GeN-Foam / GeN-Foam /



Lock

History

Find file

Web IDE



Clone



Update solvePointKineticsLiquidFuel.H

foam-for-nuclear project authored 23 hours ago

0a05c5b4



Name

Last commit

Last update

..

Make

Updated GeN-Faom to OpenFOAM v2006, w...

6 months ago

classes

Update solvePointKineticsLiquidFuel.H

23 hours ago

include

Updated GeN-Faom to OpenFOAM v2006, w...

6 months ago

main

Added optional specification of a Function1 ...

1 month ago

Allwclean

Added 1D tutorial case on boiling, uncouple...

9 months ago

Allwmake

Updated GeN-Foam with the latest FFSEuler...

1 month ago

- “Classes” contains all the physics
- “main” contains what glues them together
- “include” are folders that mainly contain chunks of code that perform specific tasks and that are included (#include) in the code

<https://gitlab.com/foam-for-nuclear/GeN-Foam/-/tree/develop/GeN-Foam>

What's inside? Tools

develop

GeN-Foam / Tools /

+

Lock

History

Find file

Web IDE

↓

↓

Clone



Resturetcured Tools folder

foam-for-nuclear project authored 8 months ago

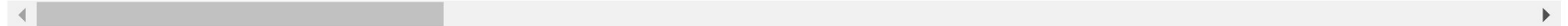
5dd726f0



Name	Last commit	Last update
..		
meshGenerationWithGmsh	Resturetcured Tools folder	8 months ago
serpentToFoam/serpent2.1.23	Resturetcured Tools folder	8 months ago
README	Resturetcured Tools folder	8 months ago

README

This folder contains helper tools that have been developed throughout the years by GeN-Foam users to simplify the us



Helper tools to make life of a user easier

- Example of a mesh creation with gmsh
- Script to convert an output of Serpent into an input for GeN-Foam

What's inside? Tutorials

develop

GeN-Foam

/ Tutorials

+ ▾

Lock

History

Find file

Web IDE

▾

📄 ▾

Clone ▾



Corrected bug in the modifiedEngel fluid-structure drag model (thanks to...



Stefan Radman authored 4 weeks ago

424e122b



Name

Last commit

Last update

..

📁 1D_HX

Corrected bug in the modifiedEngel fluid-str...

4 weeks ago

📁 1D_boiling

Updated GeN-Foam with the latest FFSEuler...

1 month ago

📁 2D_FFTF

updated regression test

1 month ago

📁 2D_MSFR

Added expected keff to Allrun.

2 months ago

📁 2D_cavityBoussinesq

Added optional specification of a Function1 ...

1 month ago

📁 2D_onePhaseAndPointKineticsCo...

Added novel feature to the pointKinetics mo...

1 month ago

📁 2D_voidMotionNoPhaseChange

Updated GeN-Foam with the latest FFSEuler...

3 months ago

- Cover essentially all functionalities of GeN-Foam
- They include a README file, an Allrun file (sometimes Allrun_parallel), an Allclean file, and some extensively commented inputs

What's inside? Documentation

develop ▾

GeN-Foam / Documentation / + ▾

Lock

History

Find file

Web IDE ▾

 ▾

Clone ▾



Deleted howTo file. Created README file in Documentation. Modified README file in tutorials.
foam-for-nuclear project authored 9 months ago

c3163d17



Name	Last commit	Last update
..		
 theory_papers	Corrected bug in pressure solution (it was n...	1 year ago
 OpenFOAMUserGuide-A4.pdf	improved documentation	1 year ago
 README	Deleted howTo file. Created README file in ...	9 months ago

README

This folder only contains reference papers and the standard OpenFOAM user guide. For information on how to install a please refer to the wiki of the gitLab repository.

Wiki

The screenshot displays the GitLab web interface. The top navigation bar includes the GitLab logo, 'Projects', 'Groups', and 'More' dropdowns, along with a search bar and utility icons. The left sidebar lists various project management tools, with 'Wiki' highlighted by a blue circle. The main area shows a file browser view of the project's wiki, listing files like 'rootCase', '.GeN-Foam', 'Allclean', 'Allrun', 'README', and 'plot.py' with their respective commit messages and timestamps. Below this, a 'README' section is visible, containing a placeholder for a description.

File	Commit Message	Time Ago
..		
rootCase	Updated GeN-Faom to OpenFOAM v2006, w...	6 months ago
.GeN-Foam	Added regression test, with missing check o...	3 months ago
Allclean	Updated tutorials	9 months ago
Allrun	Added regression test, with missing check o...	3 months ago
README	Added README, Allrun and Allclean to the 2...	9 months ago
plot.py	Added novel feature to the pointKinetics mo...	1 month ago

README

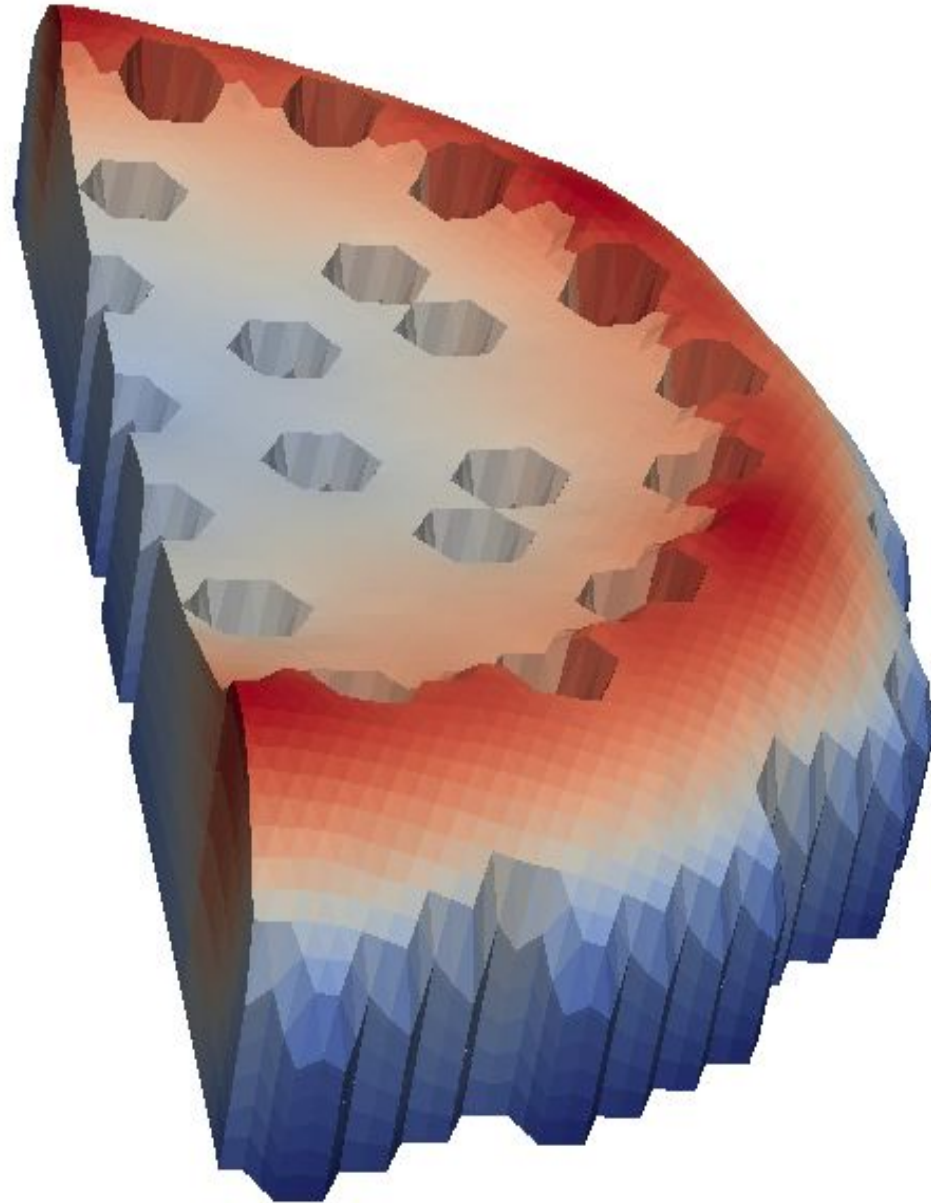
/*-----*/

DESCRIPTION

This is a placeholder for the README content.

■ <https://gitlab.com/foam-for-nuclear/GeN-Foam/-/wikis/home>

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Slack forum

foam-for-nuclear-forum

#support-run ☆
Questions and answers by the community on issues related to runtime errors and problems

August 21st, 2019

carlo.fiorina 11:10 PM
Hi Jeff. Yes, there are a few different things in the input dictionaries. You'll probably get 2 or 3 such errors when using your case on the new gen-foam

1 100 1

UNIV 1 reply View thread

carlo.fiorina 11:10 PM
You can take a look at the tutorial cases
To see how to change the dictionaries

1 100 1

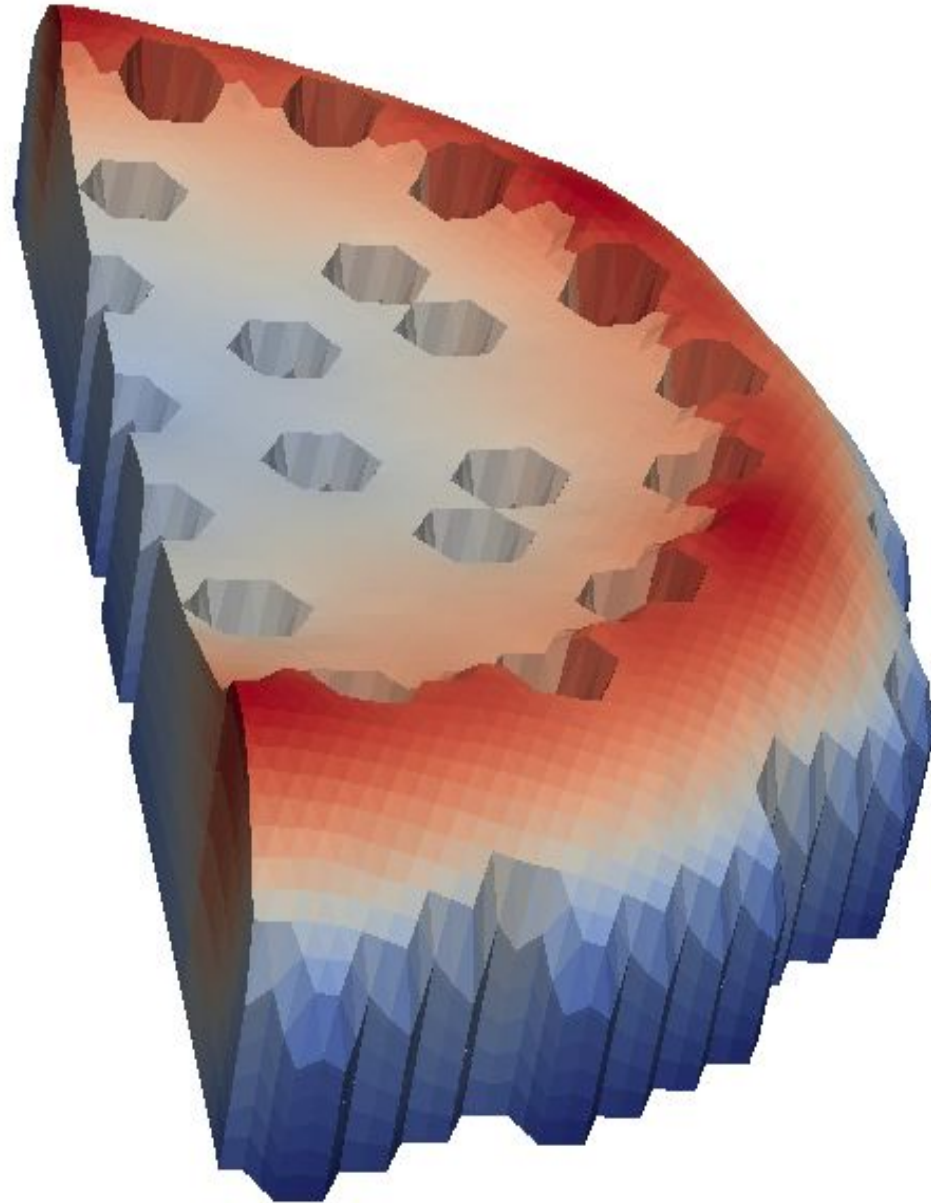
(I'm at nureth now and do not remember by heart)

Message #support-run

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- email to carlo.fiorina@epfl.ch

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An example: the tutorial

2D_onePhaseAndPointKineticsCoupling

- https://gitlab.com/foam-for-nuclear/GeN-Foam/-/tree/develop/Tutorials/2D_onePhaseAndPointKineticsCoupling
- Understanding the tutorial:
 - wiki (<https://gitlab.com/foam-for-nuclear/GeN-Foam/-/wikis/home>)
 - README file
 - Allrun file
 - Run it and use paraview to see what happens

An example: the tutorial

2D_onePhaseAndPointKineticsCoupling

- Start from the README file; do an Allrun; plot some results (https://gitlab.com/foam-for-nuclear/GeN-Foam/-/blob/develop/Tutorials/2D_onePhaseAndPointKineticsCoupling/README)

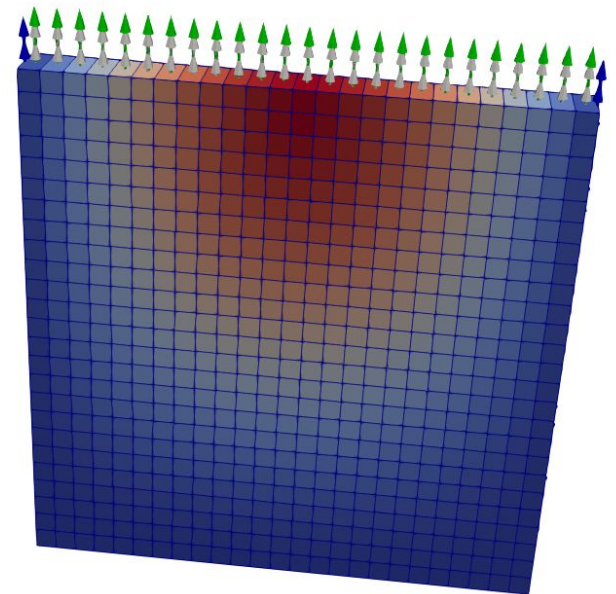
DESCRIPTION

This is a simplified test-case for the pointKinetics neutronics model. It consists of a 2-D square domain consisting of a single cellZone containing fuel. A coolant flows from the bottom of the domain to the top. Cross-sections have been arbitrarily chosen so to obtain a critical system in 2 groups.

They are used to obtain the power shape ...

Two transients are simulated:

- 0.2\$ reactivity insertion ...
- like the transient above, but including the feedback from an assumed driveline expansion ...



An example: the tutorial

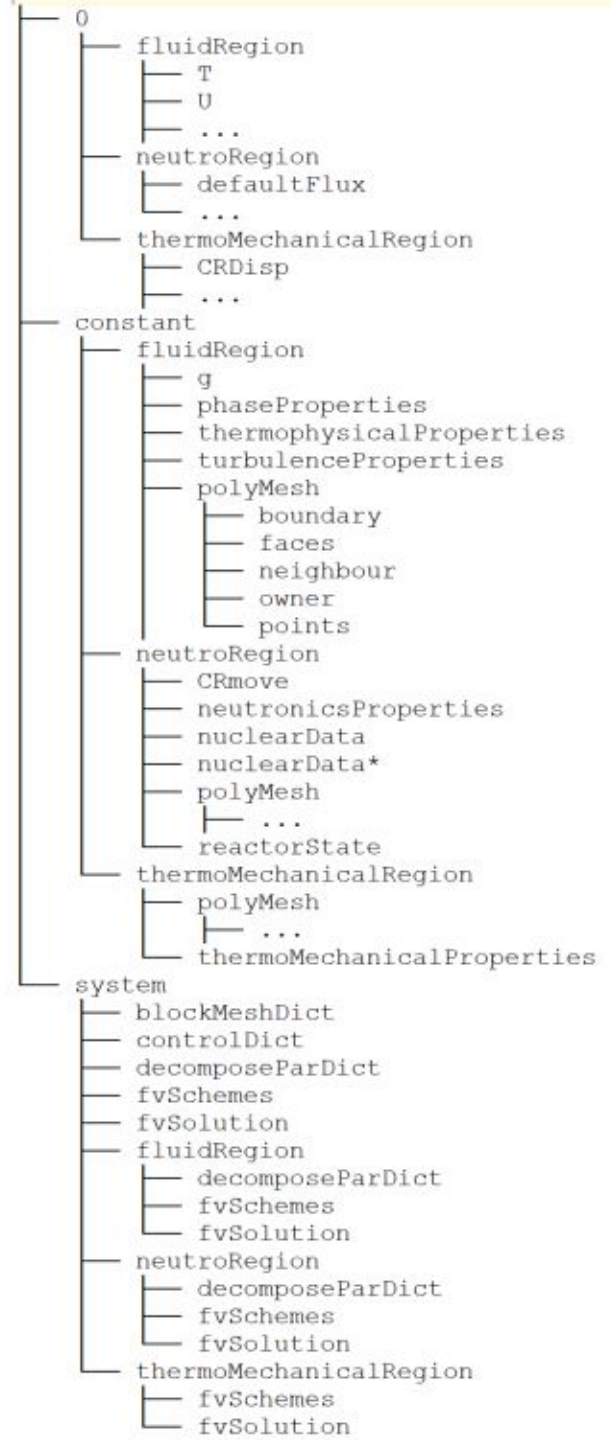
2D_onePhaseAndPointKineticsCoupling

■ Better understanding the tutorial: the Allrun file

```

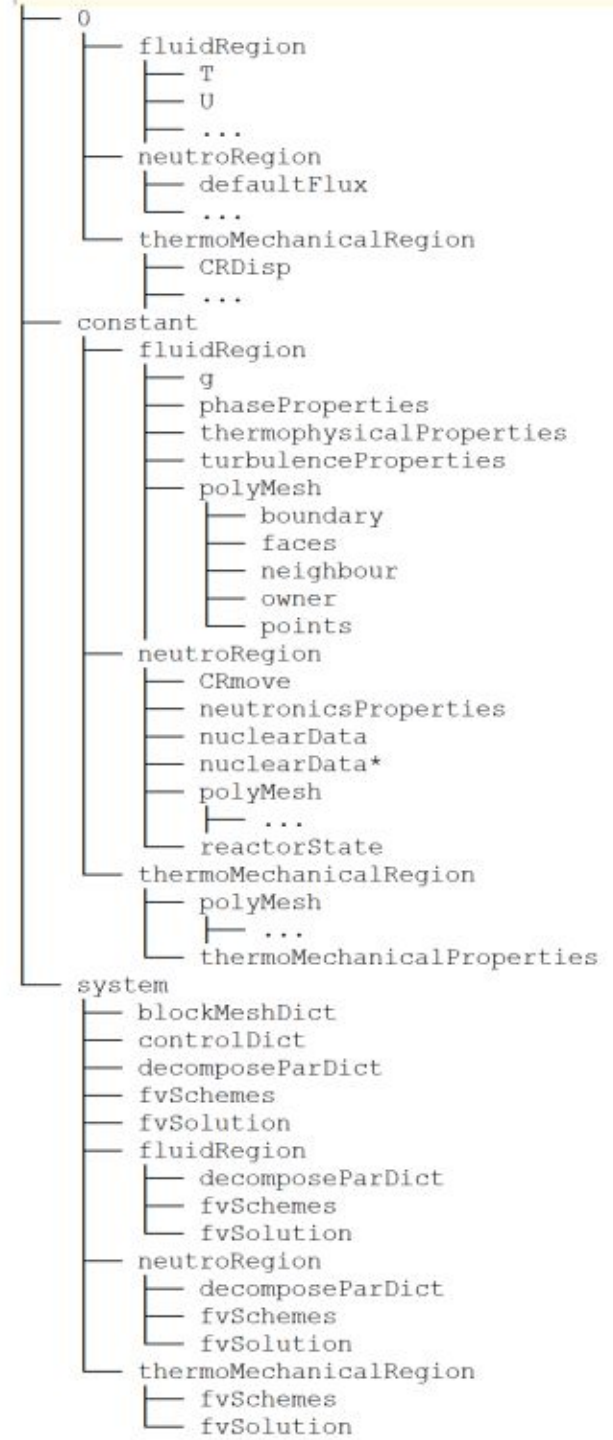
cases="steadyState transientNoDriveline transientWithDriveline"
...
setSteadyState()
{
    foamDictionary $2/constant/neutroRegion/neutronicsProperties -entry mod
-set diffusionNeutronics
    foamDictionary $2/constant/neutroRegion/neutronicsProperties -entry
eigenvalueNeutronics -set true
    foamDictionary $2/system/controlDict -entry startTime -set 0
    foamDictionary $2/system/controlDict -entry endTime -set 100
    foamDictionary $2/system/controlDict -entry deltaT -set 1
    foamDictionary $2/system/controlDict -entry writeControl -set
adjustableRunTime
    foamDictionary $2/system/controlDict -entry writeInterval -set 100
...
setTransientNoDriveline()
{
    printf "Setting transient without driveline case...\n"
    foamDictionary $2/constant/neutroRegion/neutronicsProperties -entry mod
-set pointKinetics
    foamDictionary $2/constant/neutroRegion/neutronicsProperties -entry
eigenvalueNeutronics -set false
    foamDictionary $2/system/controlDict -entry startTime -set 100
    foamDictionary $2/system/controlDict -entry endTime -set 200
    foamDictionary $2/system/controlDict -entry deltaT -set 1e-6
    foamDictionary $2/system/controlDict -entry writeControl -set timeStep
    foamDictionary $2/system/controlDict -entry writeInterval -set 10
    foamDictionary $2/constant/neutroRegion/nuclearData -entry
absoluteDrivelineExpansionCoeff -set 0.0
...

```



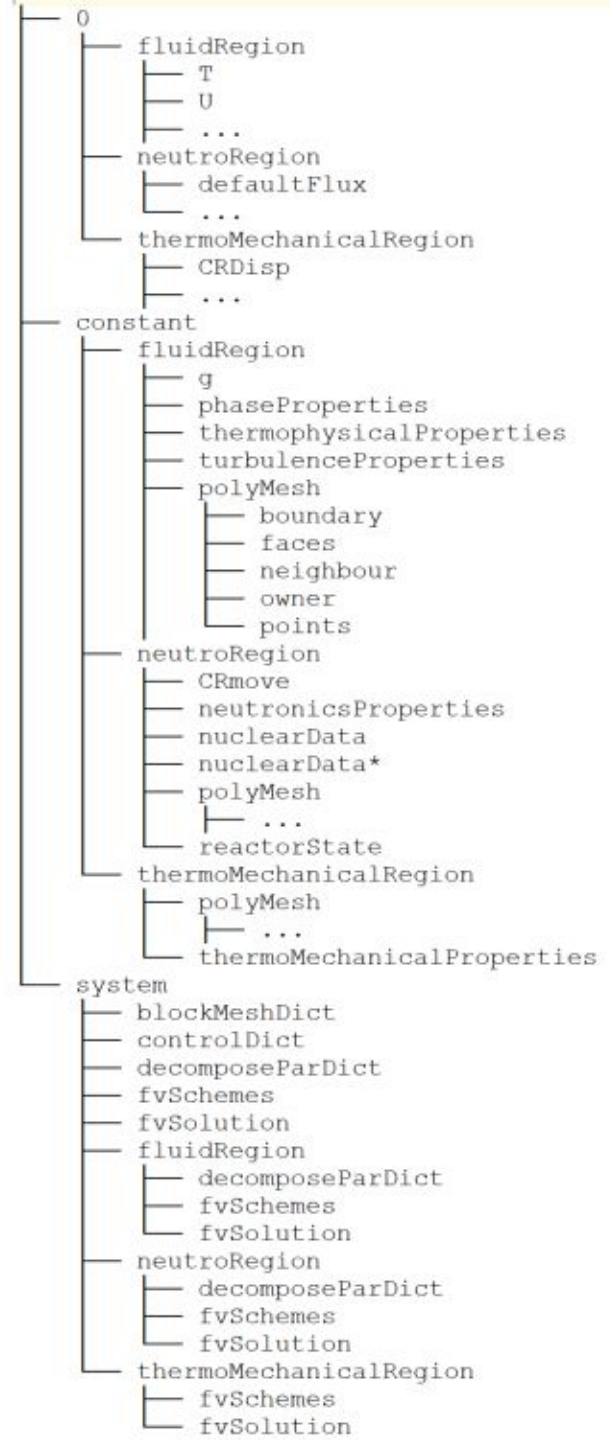
An example: the tutorial 2D_onePhaseAndPointKineticsCoupling

- Understanding the GeN-Foam case folder. 3 regions:
 - 0 folder with three subfolder containing the fields for each physics
 - *constant* folder with 3 subfolders
 - 3 meshes (*polyMesh* folders)
 - 3 sets of dictionaries
 - *system* folder with:
 - 3 subfolders with dedicated *fvScheme* and *fvSolution* for each physics
 - 1 *controlDict*
 - 1 common *fvSolution* with some multi-physics controls



An example: the tutorial 2D_onePhaseAndPointKineticsCoupling

- Better understanding the tutorial: the dictionaries
 - All of the dictionaries are extensively commented in at least one of the tutorials
 - Which tutorial to look at for every dictionary? Look in the wiki/pre-preprocessing <https://gitlab.com/foam-for-nuclear/GeN-Foam/-/wikis/Pre-processing>
 - In our case, the tutorial is mainly dedicated to the point kinetics model. Look at constant/neutroRegion/nuclearData https://gitlab.com/foam-for-nuclear/GeN-Foam/-/blob/master/Tutorials/2D_onePhaseAndPointKineticsCoupling/rootCase/constant/neutroRegion/nuclearData



An example: the tutorial

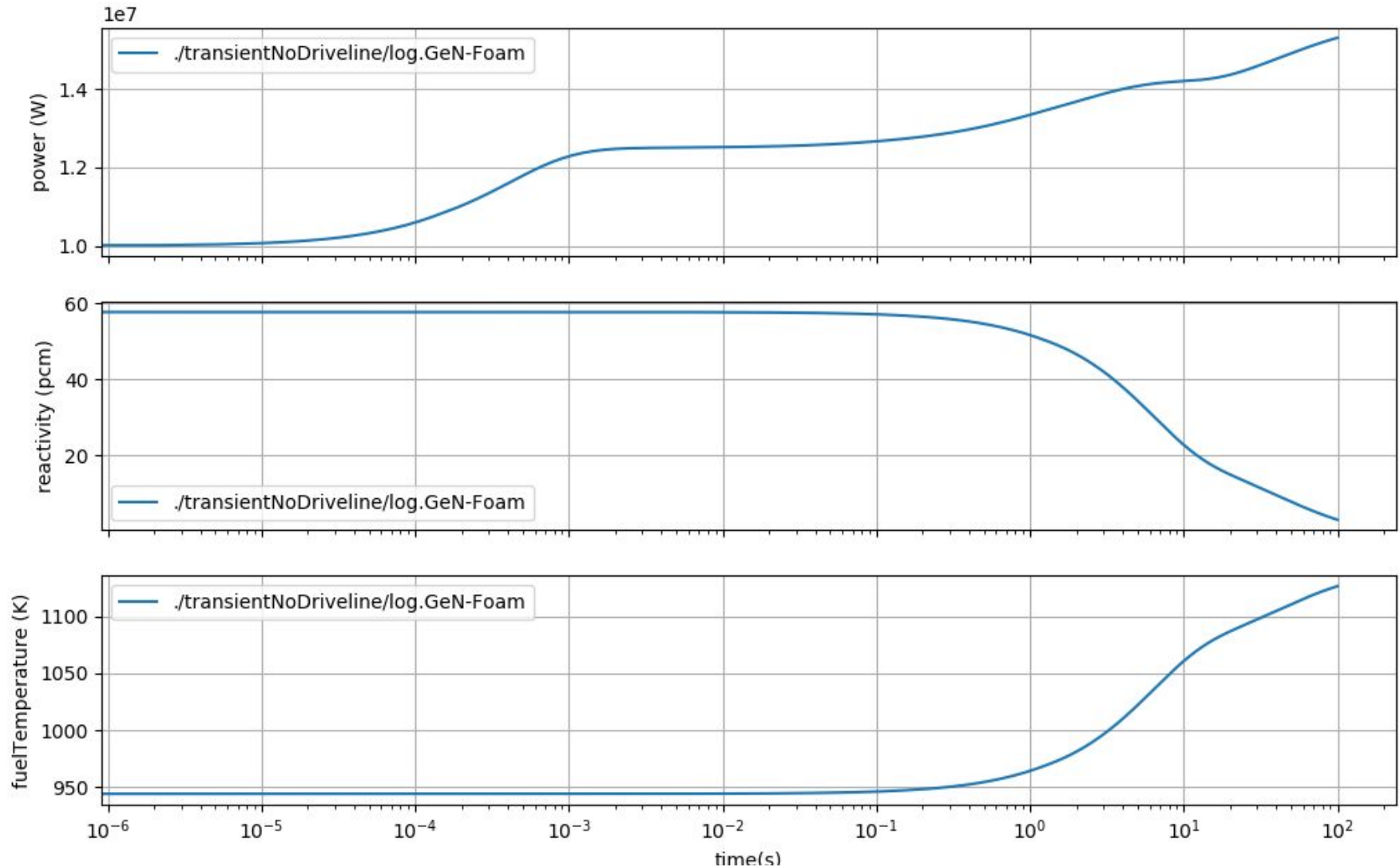
2D_onePhaseAndPointKineticsCoupling

- Run the tutorial -> ./Allrun
- Check the results:
 - Choose a folder: steadyState, transientNoDriveline, transientWithDriveline
 - Use:
 - ./log.GeN-Foam: standard OpenFOAM log
 - ./GeN-Foam.dat: quick overview of time behavior of main quantities (power, keff, min/max/average fuel and clad temp.)
 - ./constant/neutroRegion/reactorState for keff
 - in some tutorials, a python script to extract info from log file
 - paraFoam -region regionName (where regionName is set to fluidRegion, neutroRegion, or thermoMechanicalRegion)

An example: the tutorial

2D_onePhaseAndPointKineticsCoupling

- python script (extract data from log)

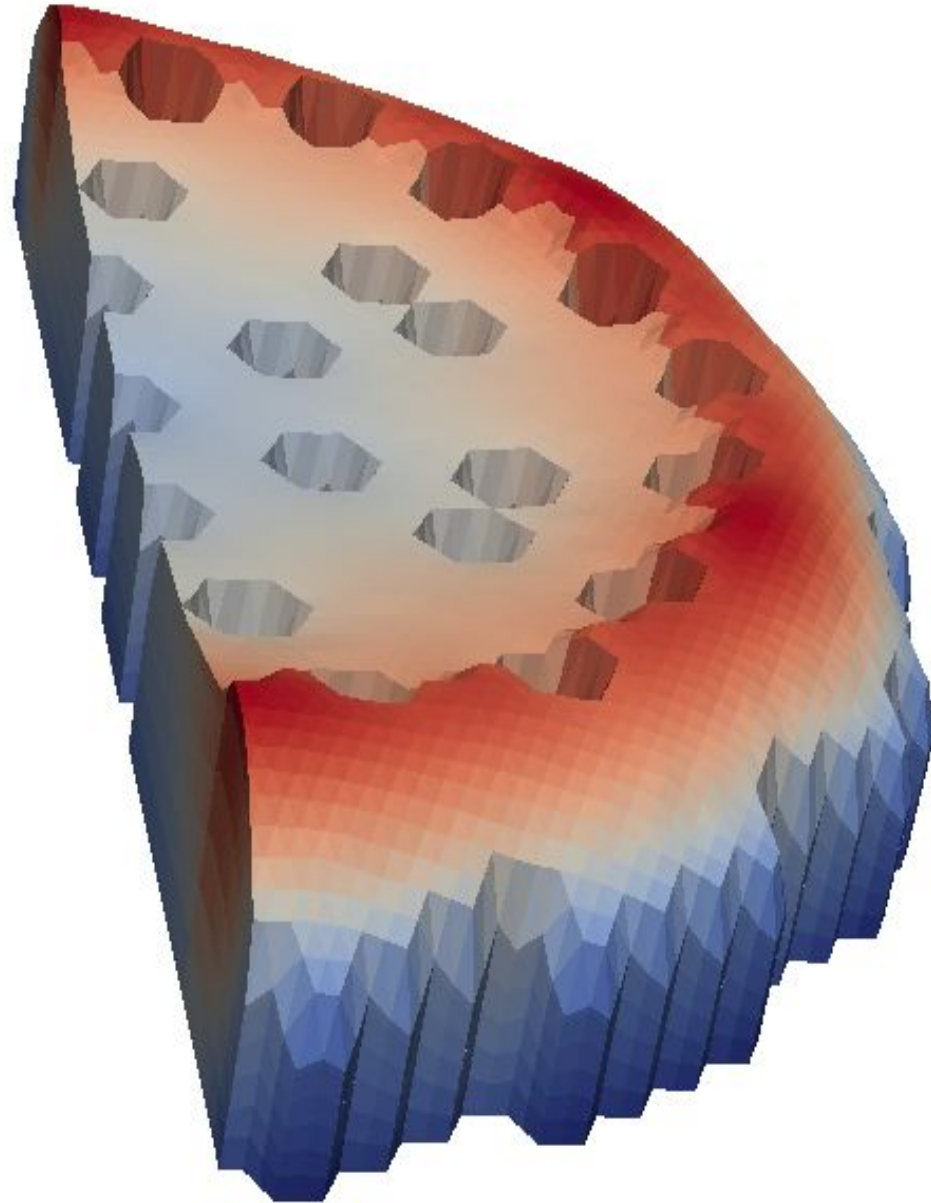


An example: the tutorial

2D_onePhaseAndPointKineticsCoupling

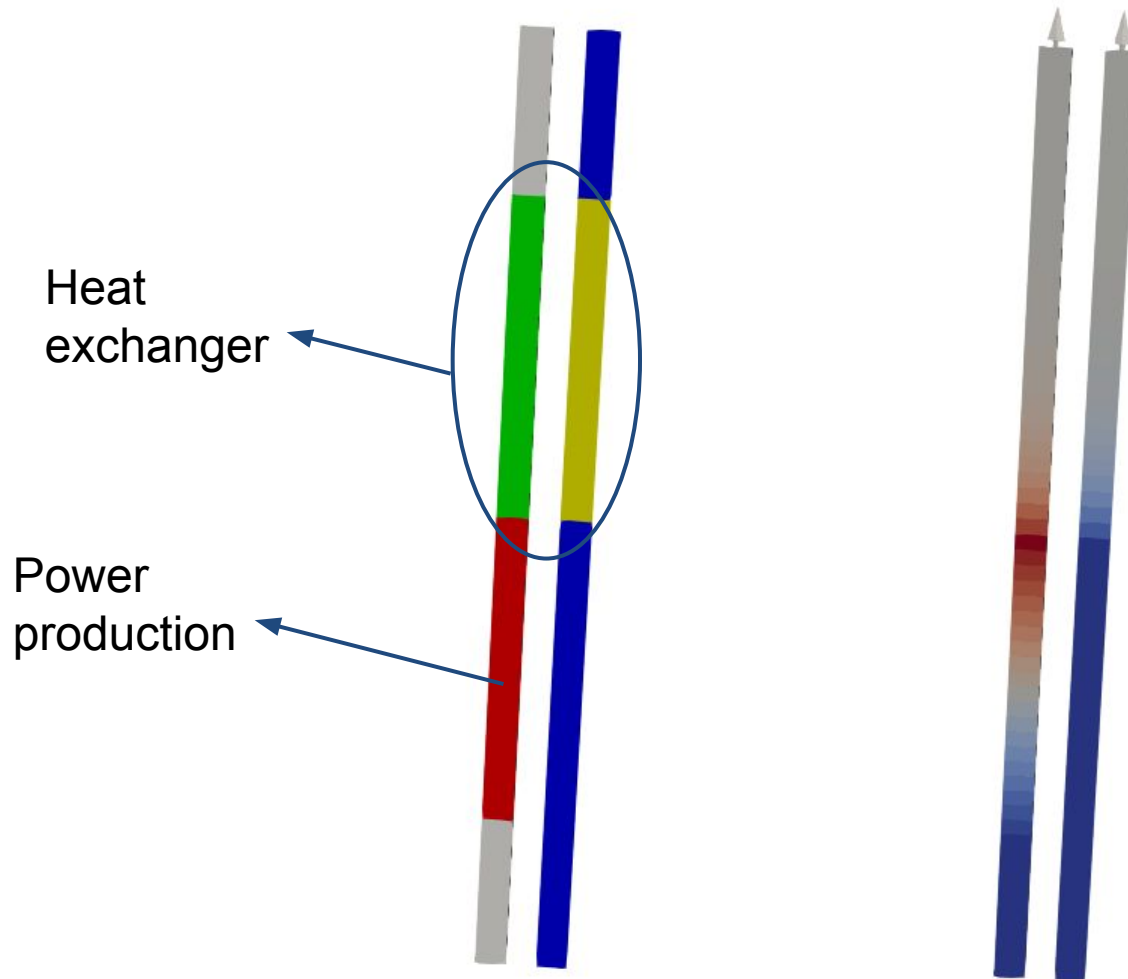
- paraview (export animation + ffmpeg)

- How to get it?
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The other tutorials: 1D_HX

- Example on how to set up a heat exchange



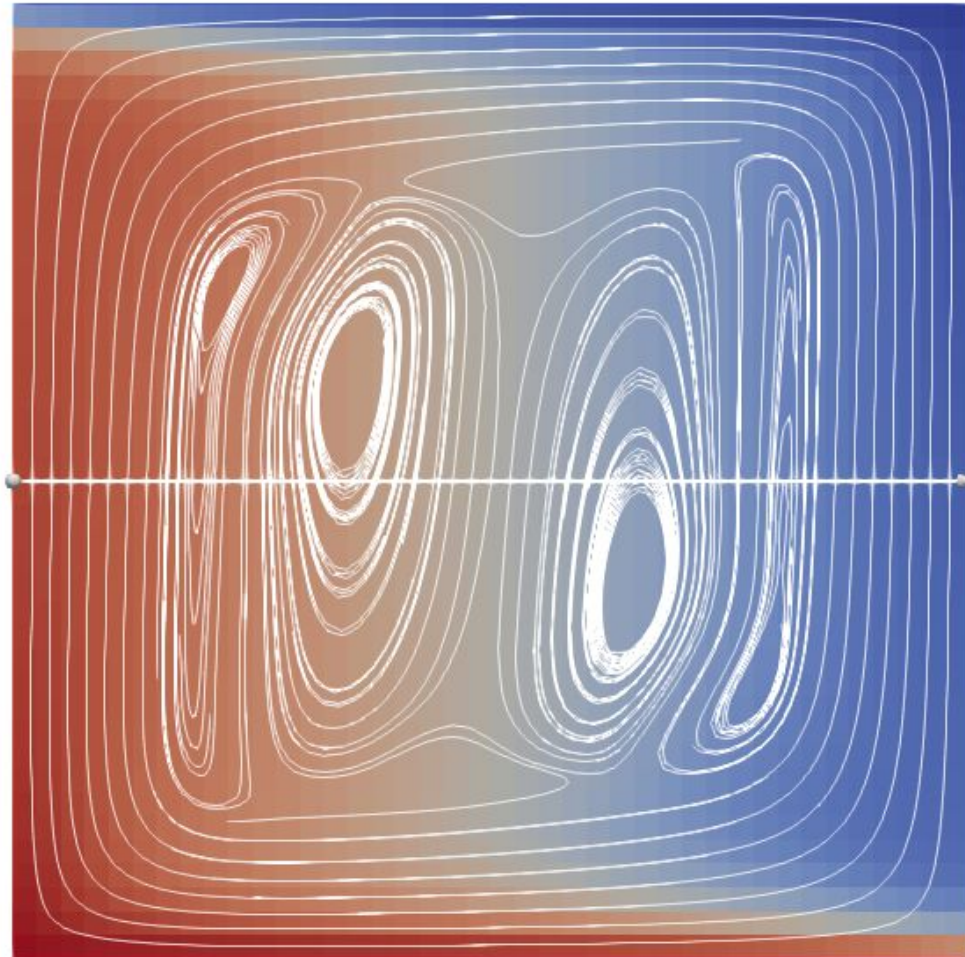
The other tutorials: 1D_boiling

- Example of two-phase simulation. 1D channel with a pressure-driven flow of liquid sodium, with power source turned on at time 0, eventually leading to boiling, After a certain time the power is turned off



The other tutorials: 2D_cavityBoussinesq

- Example of how to use of the Boussinesq approximation for buoyancy based on the standard buoyancy-driven cavity



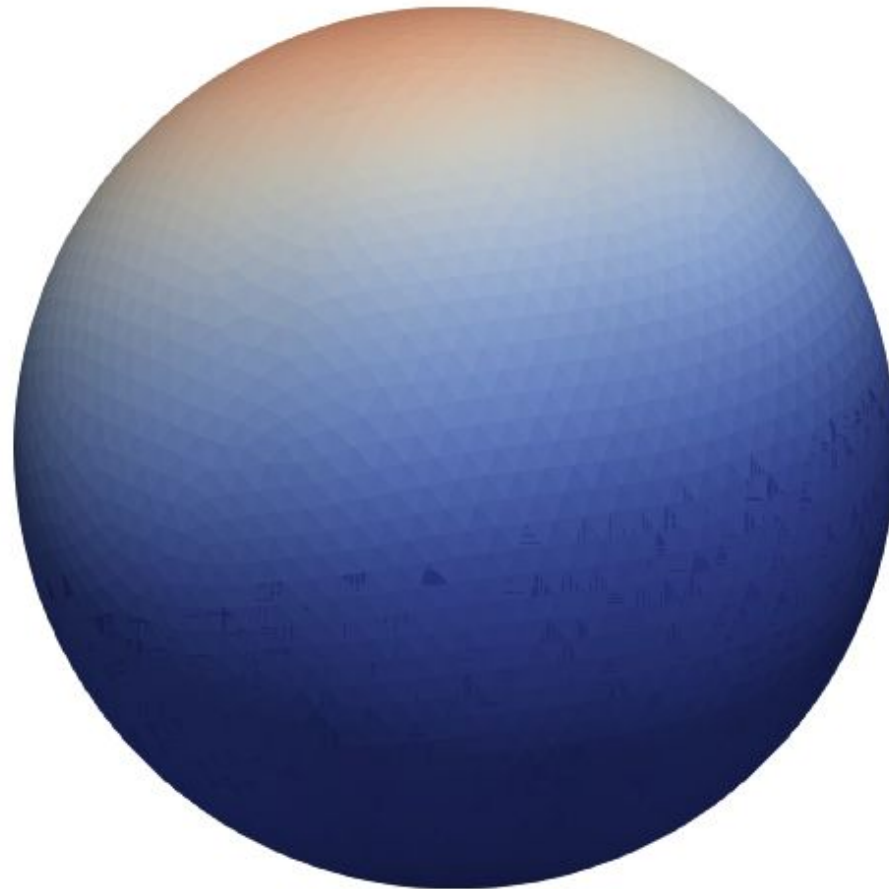
The other tutorials: 2D_voidMotionNoPhaseChange

- Simple two-phase case without mass transfer between phases



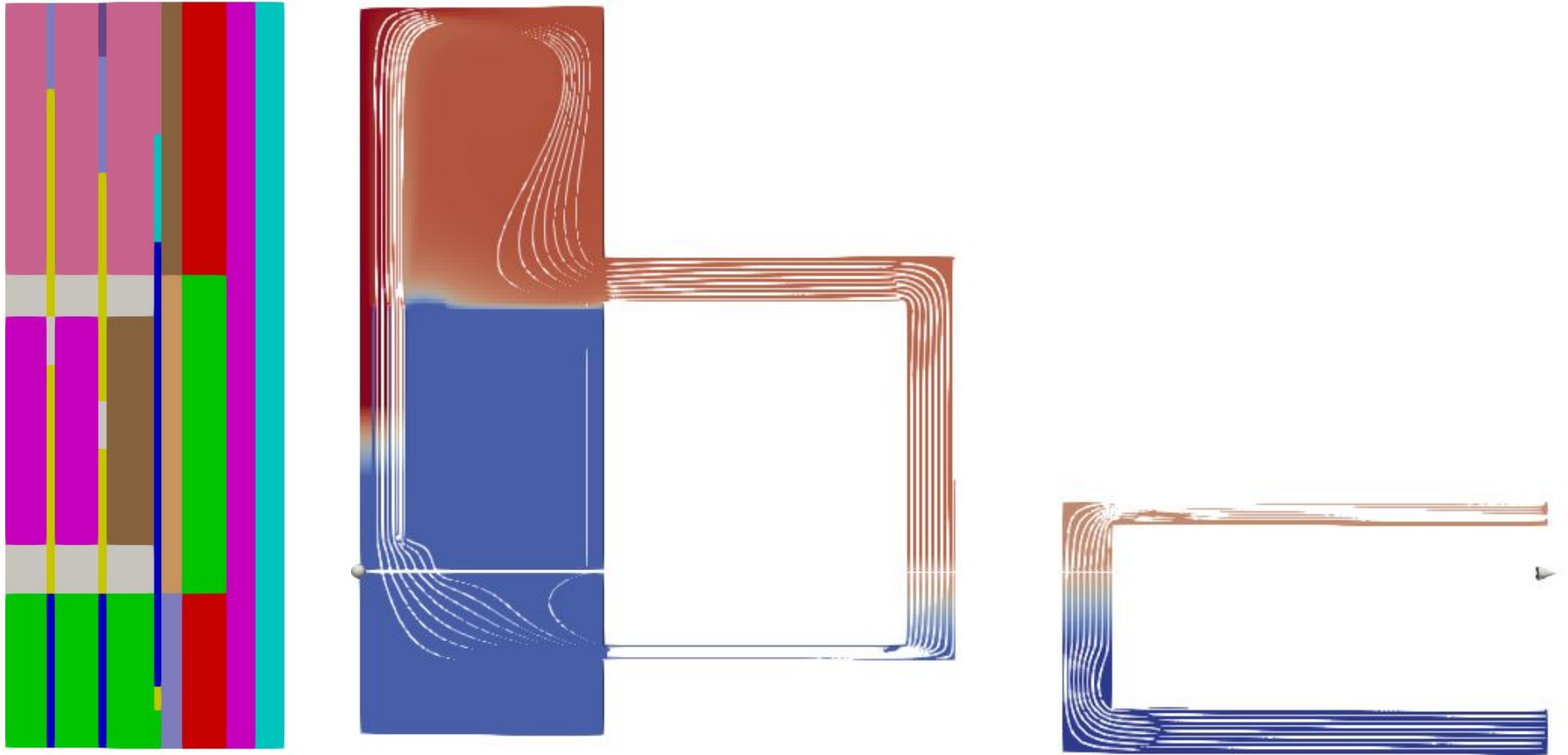
The other tutorials: Godiva_SN

- Example of a discrete ordinate calculation of Godiva



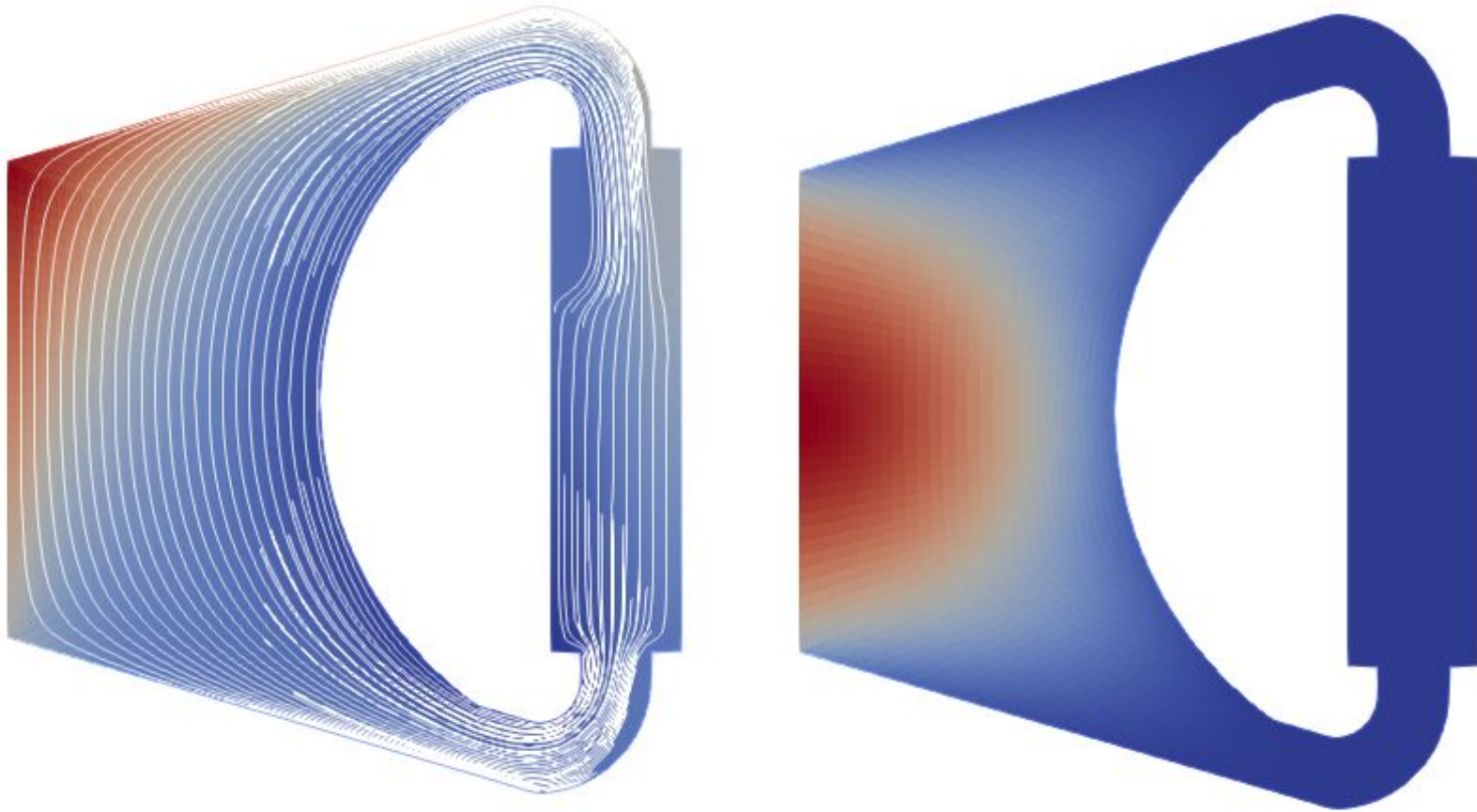
The other tutorials: 2D_FFTF

- Simplified, 2-D model of the FFTF. Simulation of a ULOF



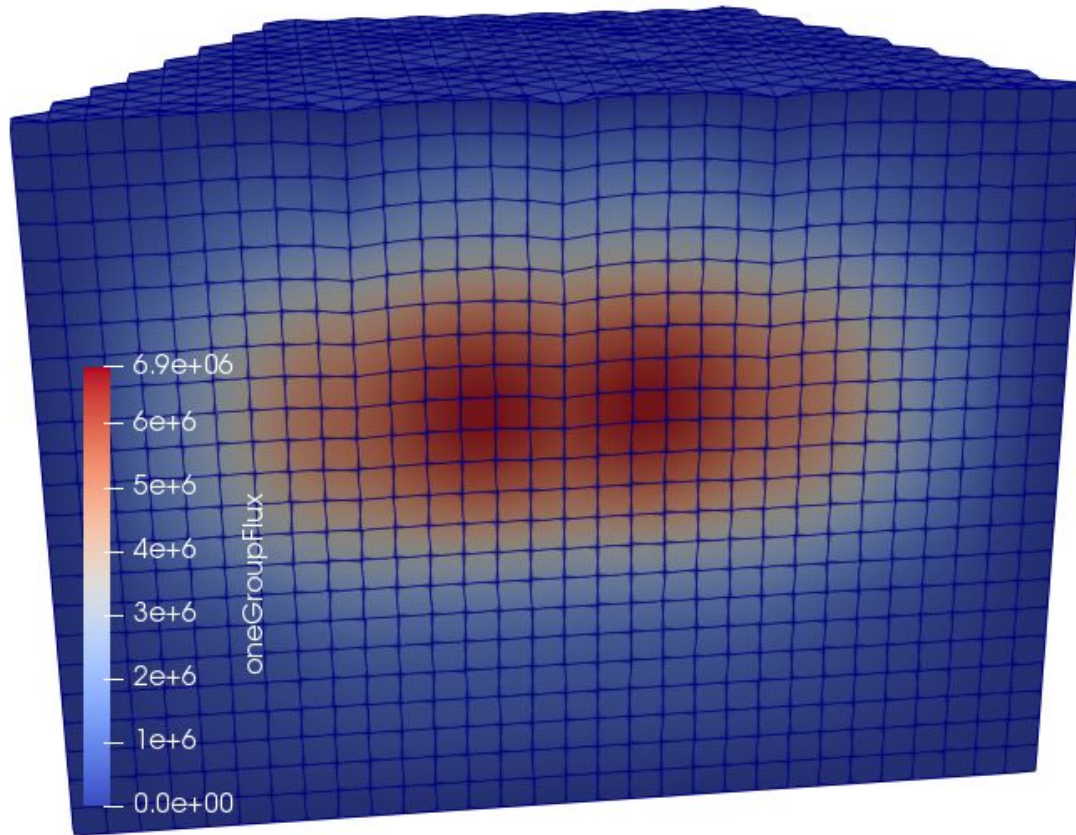
The other tutorials: 2D_MSFR

- Simplified, 2-D model of the MSFR



The other tutorials: 3D_SmallESFR

- Slightly smaller version of the European Sodium Fast Reactor
- Example of a 3D full multi-physics simulation, including core deformation



Thank you for your attention



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